

TCP/UDP and port numbers-40

Introduction

- TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are transport layer protocols in the OSI and TCP/IP models.
- Both use **port numbers** to identify applications or services on a system.
- They help direct data to the right application, even when multiple applications run on the same machine.

TCP (Transmission Control Protocol)

- **Connection-oriented protocol.**
- Establishes a connection using the **three-way handshake** before data transfer.
- Ensures reliable communication with error checking and retransmission.
- Provides sequencing so data arrives in order.
- Slower than UDP due to overhead but guarantees delivery.
- Used where accuracy and reliability matter.
- Examples:
 - Web browsing (HTTP/HTTPS)
 - File transfer (FTP, SFTP)
 - Email (SMTP, IMAP, POP3)

Transmission Control Protocol (TCP) Header

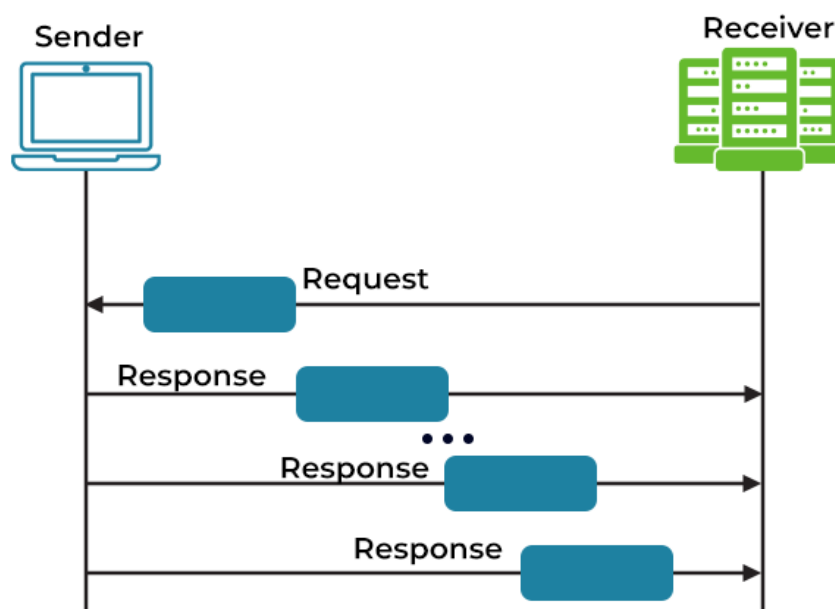
20-60 bytes

source port number 2 bytes				destination port number 2 bytes			
sequence number 4 bytes							
acknowledgement number 4 bytes							
data offset 4 bits		reserved 3 bits		control flags 9 bits		window size 2 bytes	
checksum 2 bytes				urgent pointer 2 bytes			
optional data 0-40 bytes							

UDP (User Datagram Protocol)

- **Connectionless protocol.**
- No handshake, sends data directly.
- Faster but does not guarantee delivery.
- No sequencing or retransmission.
- Suitable for real-time communication and streaming.
- Lightweight, less overhead compared to TCP.
- Examples:
 - Video streaming (Netflix, YouTube)
 - Voice over IP (VoIP)
 - Online gaming
 - DNS queries

FUNCTIONING OF USER DATAGRAM PROTOCOL (UDP)



Port Numbers

- A **port number** is a 16-bit value (0 to 65535).
- Divided into ranges:
 - **Well-known ports (0–1023)** – assigned to common services.
 - **Registered ports (1024–49151)** – assigned to user applications.
 - **Dynamic/Ephemeral ports (49152–65535)** – used temporarily for client-side communication.

Common TCP Ports

- 20, 21 – FTP (File Transfer Protocol)
- 22 – SSH (Secure Shell)
- 25 – SMTP (Simple Mail Transfer Protocol)
- 53 – DNS (Domain Name System, TCP for zone transfers)
- 80 – HTTP (Hypertext Transfer Protocol)
- 110 – POP3 (Post Office Protocol v3)
- 143 – IMAP (Internet Message Access Protocol)
- 443 – HTTPS (Secure HTTP)
- 3389 – RDP (Remote Desktop Protocol)

Common UDP Ports

- 53 – DNS (queries and responses)
- 67, 68 – DHCP (Dynamic Host Configuration Protocol)
- 69 – TFTP (Trivial File Transfer Protocol)
- 123 – NTP (Network Time Protocol)
- 161 – SNMP (Simple Network Management Protocol)
- 500 – IPSec VPN (Internet Protocol Security)
- 1701 – L2TP (Layer 2 Tunneling Protocol)

TCP vs UDP – Comparison

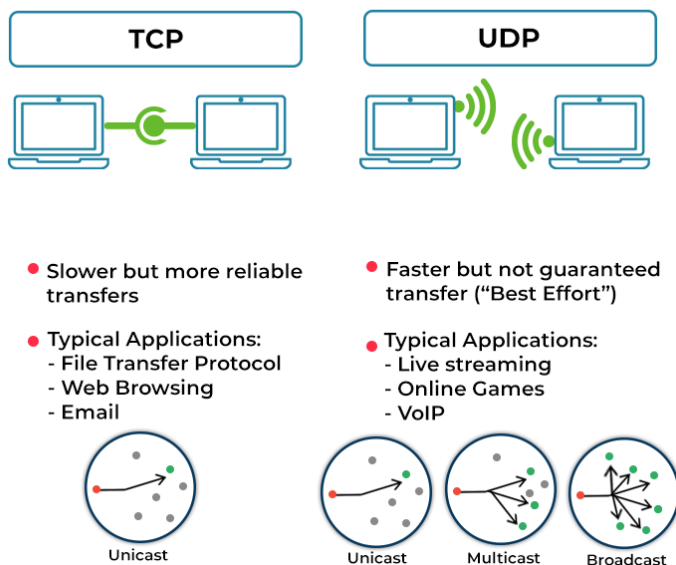
- **TCP**

- Reliable
- Connection-oriented
- Higher overhead
- Slower but accurate
- Used in file transfer, browsing, email

- **UDP**

- Unreliable (no guarantee of delivery)
- Connectionless
- Lower overhead
- Faster, best for real-time apps
- Used in streaming, gaming, VoIP

KEY DIFFERENCES BETWEEN TCP AND UDP FOR ORGANIZATIONS



Security Considerations

- Open ports are entry points for attackers.
- Unused ports should be closed or filtered by firewalls.
- Some malware exploits common ports (e.g., port 445 for SMB).
- Port scanning is often used by attackers to identify services.

- TCP and UDP are essential for transport layer communication.
- Port numbers ensure the right application receives the data.
- TCP is best for reliable communication, while UDP is ideal for fast, real-time communication.
- Proper management of ports and protocols is crucial for performance and security.